

SUSTAINABLE SMART CITY WITH IBM

Project Documentation

1. Introduction

- Project Title: SUSTAINABLE SMART CITY
- Team Leader: YOGITHA S
- Team Member: RAJESHWARI K
- Team Member: AKSHARA R
- Team Member: RITHIKA K

2. Project Overview

Purpose:

To support sustainable city development by generating eco-friendly solutions and simplifying complex policy documents.

Features:

Eco Tips Generator – Provides practical, actionable tips for sustainable living (e.g., energy saving, water conservation).

Policy Summarization – Upload a policy PDF or paste text to get clear summaries, key points, and implications.

AI Integration – Uses IBM Granite model to generate intelligent, human-like responses.

User-Friendly Interface – Gradio-based tabs for easy navigation and interaction.

Architecture

1. Libraries & Setup

Transformers, torch → Load AI model (IBM Granite) for text generation.

Gradio → Build interactive user interface.

PyPDF2 → Extract text from PDF documents.

2. Model Initialization

Load Tokenizer & Model from HuggingFace (ibm-granite/granite-3.2-2b-instruct).

Configure device usage → GPU (CUDA) if available, else CPU.

Ensure pad token setup for smooth text generation.

3. Architecture:

UI Layer – Gradio interface with Eco Tips Generator & Policy Summarization tabs.

Backend Layer – PDF text extraction, prompt creation, and response formatting.

AI Model Layer – IBM Granite LLM for generating eco tips and policy summaries.

Infrastructure Layer – Torch/CUDA for acceleration, Gradio server for hosting.

Flow: User → Gradio UI → Backend → LLM → Output

4. Setup Instructions

Prerequisites:

- Python Programming Knowledge

- Gradio Framework

-IBM Granite Model Access (via Hugging Face)

- Google Colab with T4 GPU

GitHub Account Steps:

1. Access the Naan Mudhalvan Smart Internz portal.

2. Choose an IBM Granite model from Hugging Face.

3. Run the application in Google Colab with required libraries.

4. Upload final project files to GitHub

5. Folder structure

Project_root/

App.py → Main Gradio application

Models/

__init__.py

Model_loader.py → Load IBM Granite model & tokenizer

Utils/

Pdf_utils.py → Extract text from PDFs

Prompt_utils.py → Prompt creation & response cleaning

Static/ → (Optional) Store images, logos, UI assets

Requirements.txt → Dependencies (gradio, torch, transformers, PyPDF2)

README.md → Project overview & usage instructions

Data/ (Optional) → Sample PDFs or text files for testing

6. Running the Application

1. Open Google Colab and load the project notebook.
2. Install dependencies and configure runtime with GPU.
3. Run the notebook cells to start the Gradio app.
4. Access the provided link to interact with Sustainable Smart City.

7. API Documentation

Sustainable Smart City provides endpoints for:

- Asking questions about government services.
- Uploading feedback for sentiment analysis.
- Viewing summarized policies.

- Accessing dashboards and reports.

8. Authentication

- For the demo, Sustainable Smart City runs in an open setup. In real deployments, authentication methods like API keys, OAuth2, and role-based access would be used.

9. User Interface

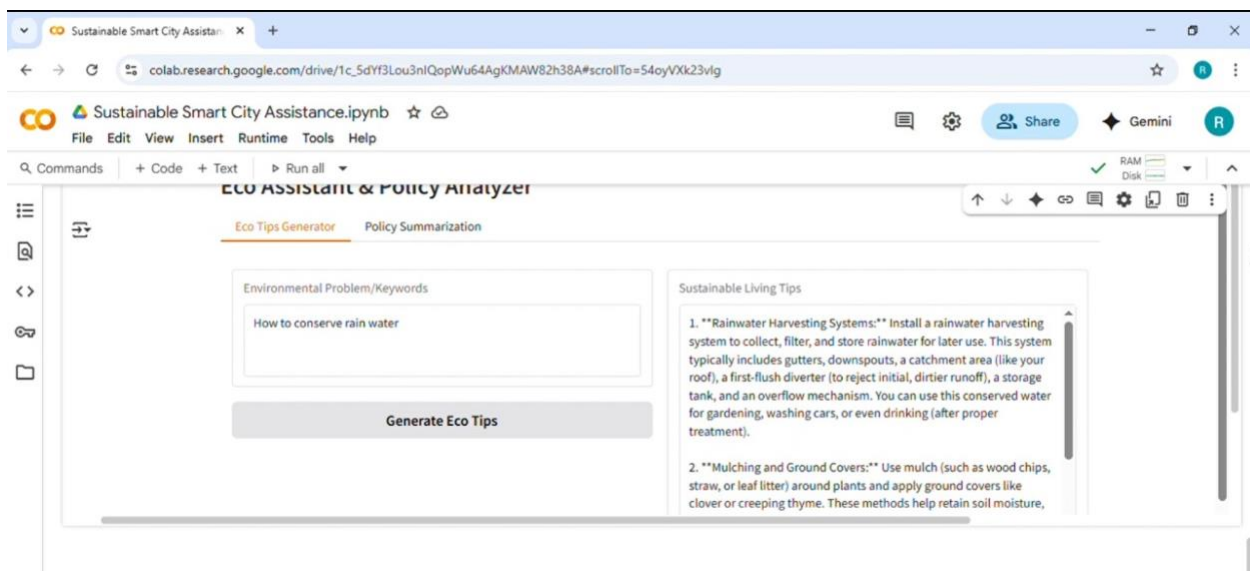
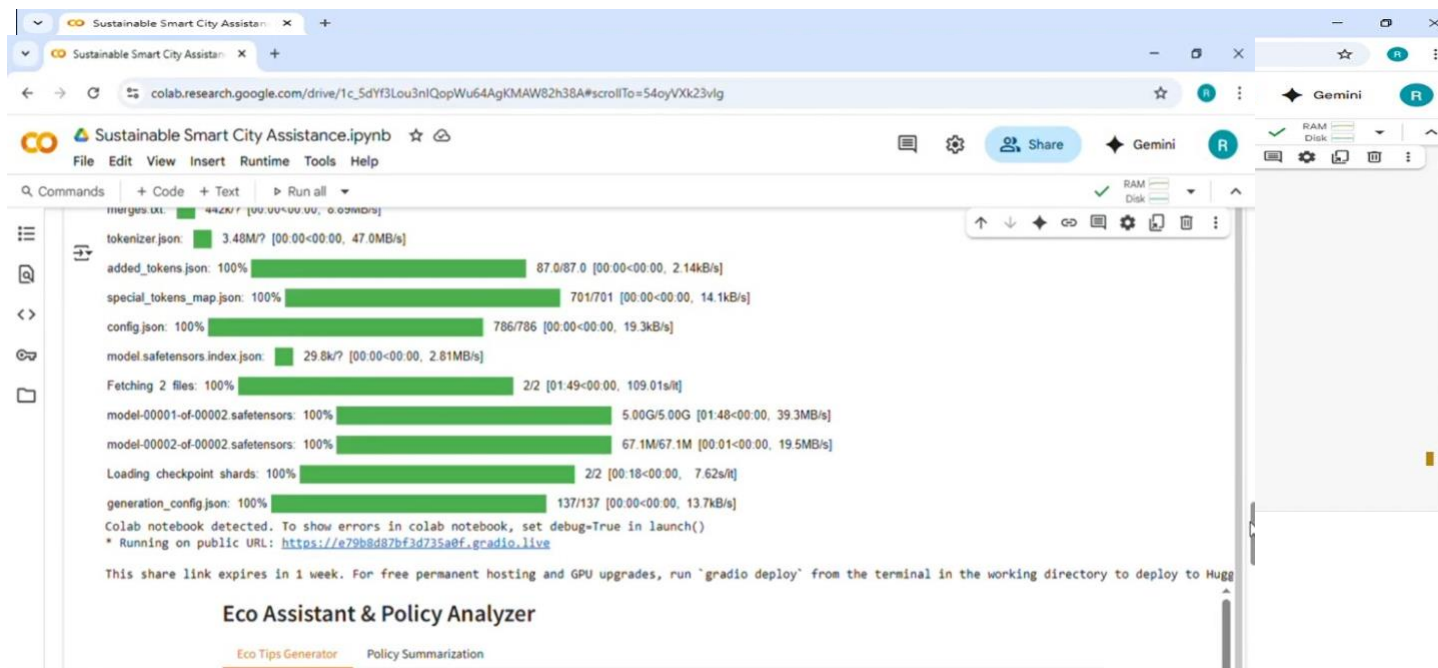
- The Gradio interface is clean and simple with:
 - A chatbox for queries.
 - Dashboard views for officials.
 - Options to upload feedback and documents.

10. Testing

Testing included:

- Unit testing for AI responses.
- Manual testing of the Gradio interface.
- Edge case handling with unexpected inputs.

11. Screenshots



12. Known Issues

Promotes eco-friendly lifestyle.

Summarizes sustainability policies.

Supports education & awareness.

Helps city planners with quick insights.

13.Future Enhancements

IoT & real-time data integration.

Carbon footprint calculator.

Multilingual & mobile app support.

Cloud deployment with secure API keys